

Harikrishnan S

PhD Scholar, IIT Palakkad

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Research Scholar specializing in Power Electronics and Drives, with research focused on encoderless speed control of electric drives. Experienced in the design, development, and experimental validation of hardware platforms, with expertise in real-time control, speed estimation, digital implementation, and embedded control systems. Skilled in translating theoretical control concepts into practical hardware implementations, complemented by experience in software and application development for engineering and research applications

Education

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| 2022 – Present | ■ PhD in Electrical Engineering
at Indian Institute of Technology (IIT), Palakkad; CGPA: 10/10
Guide: Dr. Arun Rahul S (Assoc. Professor) |
| 2022 – 2024 | ■ M.Tech in Power Electronics and Power Systems
at Indian Institute of Technology (IIT), Palakkad; CGPA: 9.84/10 |
| 2016 – 2020 | ■ B.Tech, in Electrical and Electronics Engineering,
at Thangal Kunju Musaliar College of Engineering (TKMCE), Kollam
under APJ Abdul Kalam Technological University, Kerala; CGPA: 9.5/10 |
| 2013 – 2015 | ■ Class XII
at Trinity Lyceum, Kollam (ISC); Aggregate: 97.25% |
| 2012 – 2013 | ■ Class X
at Trinity Lyceum, Kollam (ICSE); Aggregate: 94% |

Skills

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| Coding | ■ Java, xml, Python, MATLAB, C, HTML, ... |
| Hardware | ■ TMS320F2837xD, Arduino, NodeMCU, PCB Design, Renesas RX23T, |
| Software | ■ Code Composer studio, Android studio, Latex, PLECS, IPE, Adobe illustrator, Proteus, Power World, KiCAD, MATLAB, PSCAD, ETap, Jetbrain, Inkscape, Qspice, Ltspice, CS+... |
| Misc. | ■ Academic research, Teaching, Training. |
| Languages | ■ Strong reading, writing and speaking competencies in English. |

Work Experience

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| Jul 2021 – Aug 2022 | ■ Tutor , Info On Learning Solutions .
Worked as a tutor for Info On Learning Solutions, managing tuition sessions for multiple students across various subjects in the Electrical and Electronics domain. Responsible for delivering clear, effective instruction and supporting students' understanding of key concepts. | Kerala, India |
| Jan 2025 – April 2025 | ■ Project Assistant , IIT Palakkad.
Assisted Dr Arun Rahul S, in development of a encordless speed control scheme for PMSM centrifugal pump for CRI Pumps. Proposed alogrithm was implemented in Renesas Rx23T microcontroller and field tested successfully for a 3HP and 5HP centrifugal pump. | Kerala, India |

Research/Academic Experience

Aug 2022 – Present

■ Teaching Assistant, IIT Palakkad.

Kerala, India

- EE5024 DC-DC Power converters Analysis and Simulation (PG level) : Assisted faculty in formulation and conduct of simulation experiments in Qspice, lab evaluation and viva voce examinations
- EE5103 Power Converter Design Lab (PG level): Assisted faculty in experiment design, preparation of lab manuals, lab demonstrations, and conducting viva voce examinations.
- EE2020A Electrical Machines (UG level): Assisted faculty in preparation of tutorials, assignment evaluations, lab demonstrations, and viva voce examinations.

■ Workshop Instructor

- Master Trainer for Solar Ambassador Workshop (Energy Swaraj Foundation). Conducted a 1-day workshop at IIT Palakkad on Oct. 2, 2024. Kerala, India
- Conducted a 2-day hands-on workshop on App Development (Online) under EEE Association, TKMCE. Kerala, India
- Conducted a 2-day workshop on Circuit Design and Arduino under IEI, TKMCE. Kerala, India

Projects (selected)

- **Low-Cost Microcontroller-Based Real-Time Digital-Twin-Enabled HIL Toolchain for Model-Driven Simulation, Embedded Deployment and Data Acquisition** Aug. 2025 – 2026
- **State Feedback Control of PMDC Motor Drive with Current-Constrained Speed Regulation** Aug. 2023 – 2024
- **Enhancement of Microscope Vision via Fourier Ptychography Imaging for Lung Cancer Detection** Aug. 2019 – 2020
- **IoT Module for Implementation of IoT-Based Switching in Common Household Wiring** Aug. 2018 – Mar. 2019
- **Android Applications: Published** Aug. 2022 – Present
 - EEE Buzz, LCS App, Attendcal, QuarmTKM

Research Publications

- Harikrishnan S and Arun Rahul S, "A Practical Design Methodology for Cascaded PI Control of Electric Drives Under Saturation Constraints" in *IECON 2026, Qatar* (accepted)
- Harikrishnan S and Arun Rahul S, "Analysis of Modified Single Loop PI Control Framework for Electric Drive Applications" in *PEDES 2026, India* (accepted)
- Harikrishnan S and Arun Rahul S, "A Generic Online Resistance Adaptation Scheme for Speed Observers in Encoderless DC Motor Drives" in *PEDES 2026, India* (accepted)
- Hariharan D, Radhika Gopal, Harikrishnan S and Arun Rahul S, "Implementation-Oriented Evaluation of PMDC Drive Sensorless Speed Control Techniques" in *PEDES 2026, India* (accepted)
- Harikrishnan S and Arun Rahul S, "Application and Comparative Study of the $\alpha\beta$ Filter in Model-Based Speed Estimation and Sensorless Control of PMSMs", in *NPEC 2025, India*. **(Best Paper award)**

Research Publications (continued)

- Arun Rahul S and Harikrishnan S, "Closed Loop Power Hardware in the Loop Back EMF and Current Emulation of a PMDC Machine", in *ECCE-Asia 2025, India*
- Harikrishnan S and Arun Rahul S, "State Feedback Speed Control of PMSM Using Multithreaded Controller", in *PEDES 2024, India*.
- Arun Rahul S and Harikrishnan S, "Speed Sensorless Power Hardware In the Loop Emulation of PMDC Machine with Fuzzy Based Observer", in *PEDES 2024, India*.
- Harikrishnan S and Arun Rahul S, "Multithreaded State Feedback Control with Speed Estimation for Permanent Magnet DC Motor Speed Control", in *RAICS 2024, India*.
- Harikrishnan S and Arun Rahul S, "Energy efficiency-based Multithreaded State Feedback Speed Control of PMDC Motors", in *IAES 2024, India*
- Sikhin VC and Harikrishnan S, "Human detection system for rescue operations", in *ICMSS'19, India*

Positions of Responsibility

- **Student Editor** 15th volume of magazine "Potentia", Annual Technical Magazine of EEE dept, TKMCE Oct 2020 - Dec 2020
- **Head volunteer for RECCAP 2026, IEEE conference** May 2026
- **IIT Palakkad Science Quest EE Dept. Co-ordinator** May 2025, May 2026
- **IIT Palakkad Open house EE Dept. Co-ordinator** Jan 2024, Jan 2025
- **Research Scholar's Day Program Convenor** Feb 2024, Feb 2026
- **Technical Head** - IEI, Electrical student chapter TKMCE Aug 2018 - Mar 2019

Achievements

- **B-Tech University Rank Holder** Graduated with Honours in Electrical and Electronics Engineering under APJ Abdul Kalam Technological University, Kerala with university rank of 8 in 2020.
- **District Level finalist of the NASA Space App Challenge**, Worked as an App developer under the team "Star Trek", who went on to win the Kerala Kollam Zone NASA Space App Challenge.
- **Developed an android app "QuarmTKM"** Developed an android application in association with the Kollam district administration by the name "QuarmTKM" to monitor people in quarantine during the pandemic
- **Best Paper award** at NPEC 2025.
- **Best Poster award** at IAC 7.0, 2025 held at IIT Palakkad
- **Won multiple prizes in various Technical competitions at all Kerala level.**

References

Dr.Arun Rahul (PhD Supervisor),
Assoc. Professor at IIT Palakkad,
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